# User guide of

# EMV kernel library API

# (V2.4.104)

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Urovo Technology Co., Ltd.

# Revision declaration

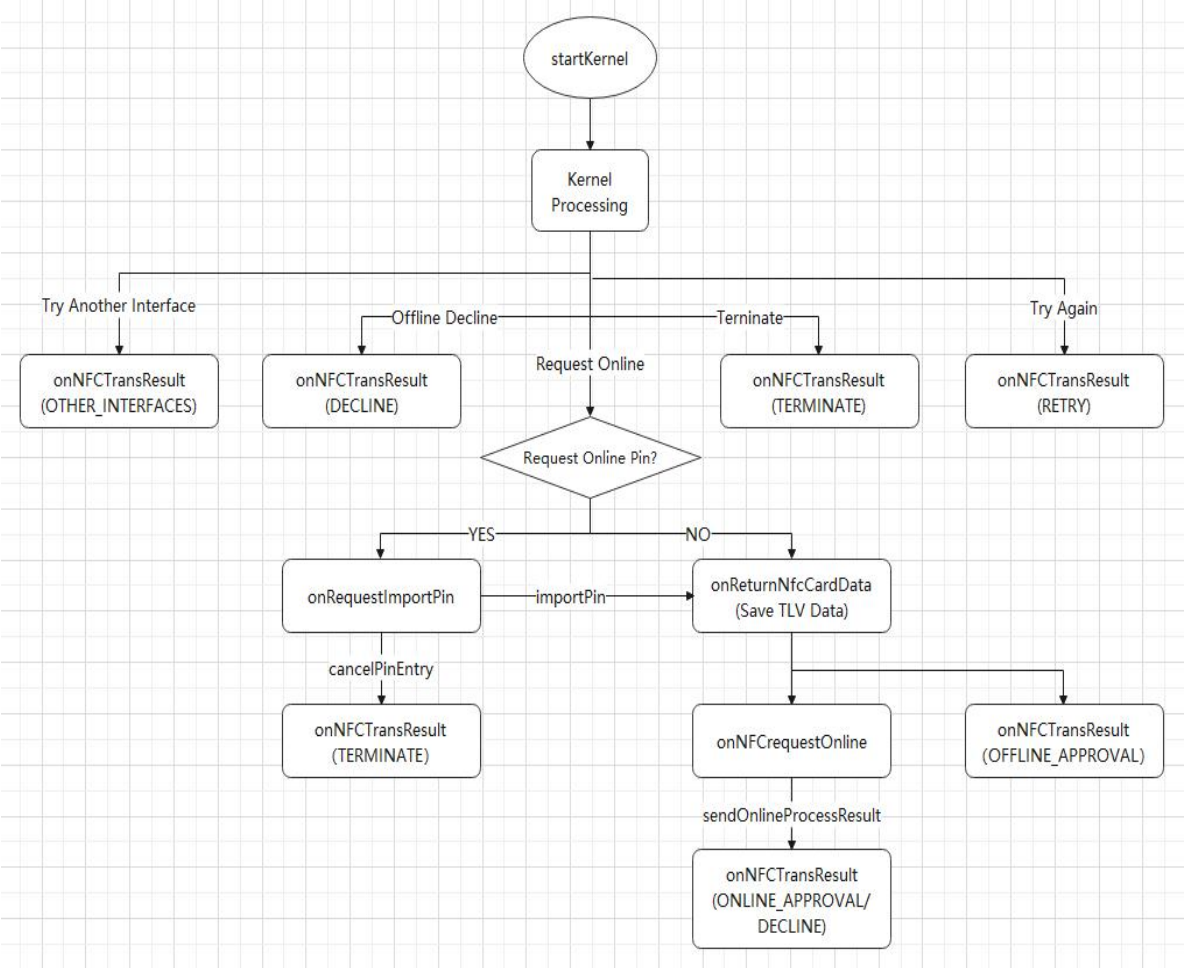
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# Overview

This is EMV SDK interface Guide, This document provides the guideline on how to integrate UROVO POS devices into mobile payment application (or App) to accept either magnetic or EMV chip cards.

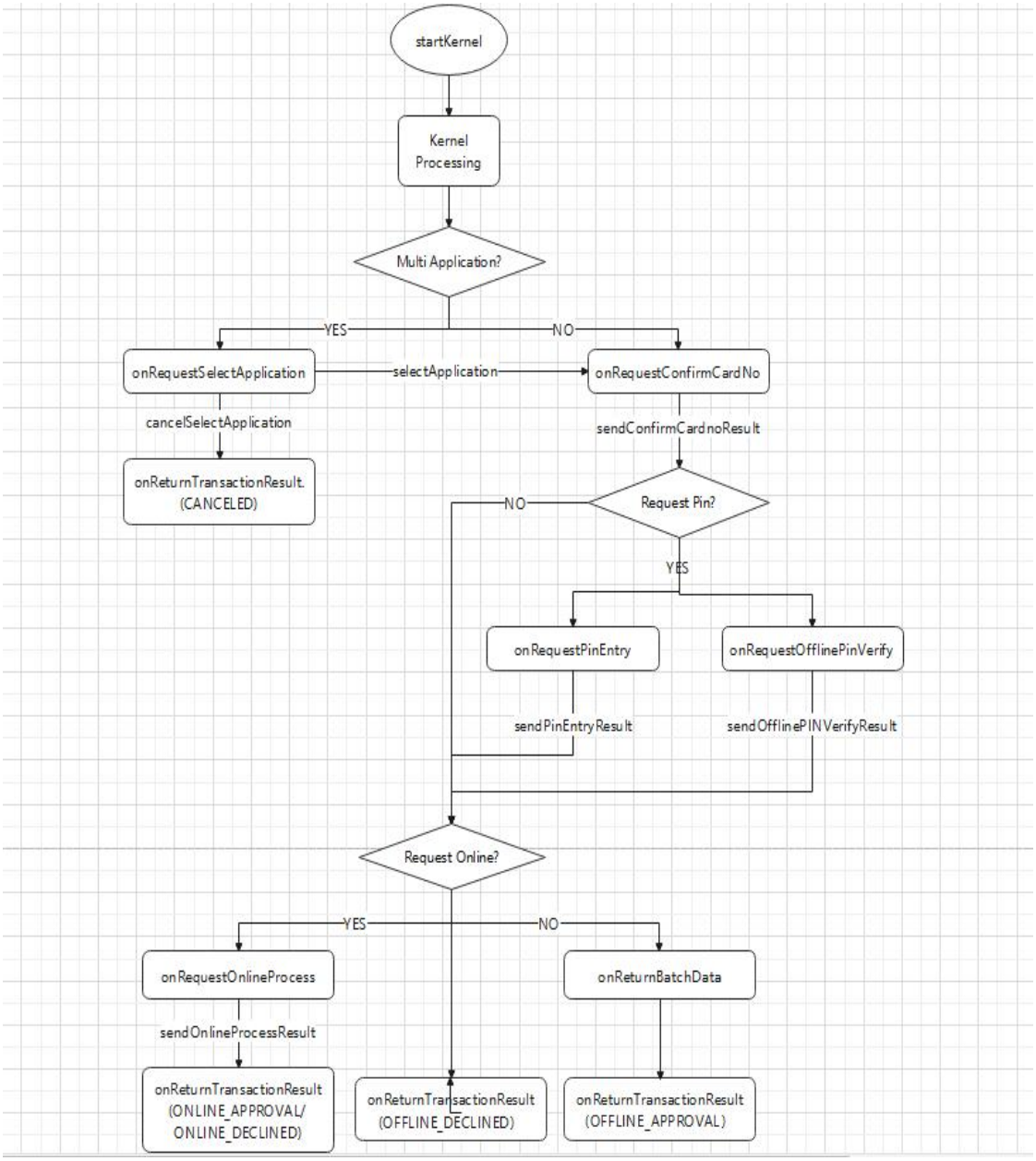
**If you want check other API, like printer scanner, please reference Platform SDK <UrovoPosSdk>**

## NFC Transaction Flow




**Description:** **call startKernel start the contactless transaction ,if card need CVM ,eg online pin , will callback onNFCrequestImportPin , then if input sussessed , you should call mKernelApi.sendPinEntry(); then kernel will continue transaction , will callback onReturnNfcCardData , you should save the online data or batch data in this callback , if kernel request online , will callback onNFCrequestOnline , you should send online messages to host in this callback , when your host return online response , you should call mKernelApi.sendOnlineProcessResult(true, responseData); then kernel will callback onNFCTransResult ,tell you the final transaction result**

## EMV Transaction Flow




**Description: the first call startKernel start the contact transaction , kernel will check card according to checkCardMode , if card have mult app , will call back onRequestSelectApplication , you should call selectApplication continue the transaction , then kernel will call back onRequestConfirmCardno to verify the cardno , you should call sendConfirmCardnoResult(true); continue the transaction , if need CVM , eg :online pin , onRequestPinEntry will callback , you should start pinpad , then if input sussessed , you should call mKernelApi.sendPinEntry(); continue the transaction , then the kernel will callback the**

**onRequestFinalConfirm() , Request final confirmation , you should call mKernelApi.sendFinalConfirmResult(true); then kernel will execute the Terminal Behavior Analysis , if the transaction need online ,will callback onRequestOnlineProcess , you should send online data(DE55 or other tag you need) to host in this callback ,when your host return online response , you should call mKernelApi.sendOnlineProcessResult(true, responseData); then kernel will execute the second GAC , callback the onReturnTransactionResult , tell you the final transaction result**

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# 1.API

## 1.01 startKernel

| Declare： | **void startKernel(Hashtable<String,Object> transParams)** |
|---|---|
| Description： | Start the transaction with specified CheckCardMode,go through all the EMV process. |
| Parameters： | See below Hashtable keys table. |
| Return： | None |
| Remark： | onRequestSetAmount |

startEmv Hashtable Keys 1/2

| Transaction Settings |
|---|
| • **emvOption**: Indicate whether to force transaction go online, see *EmvOption* enumeration.<br>• **checkCardMode**: Indicate which check card mode to use, see *CheckCardMode* enumeration.<br>• **transactionType**: Indicate the type of financial transaction (eg:00—goods, 01—cash ,09—cashback, 20--refund)<br>• **amount**: The transaction amount. (e.g. “0.01”)<br>• **cashbackAmount**: The cashback amount.<br>• **amountEx**: The transaction amount. (e.g. “001”)<br>• **cashbackAmountEx**: The cashback amount.<br>• **currencyCode**: The 3-digits transaction currency code (e.g. “840” for USD).<br>• **checkCardTimeout**: Indicate the check card timeout, in second.<br>• **isEnterAmtAfterReadRecord:** this flag is true, you need to enter the amount in the onRequestSetAmount callback<br>• **FallbackSwitch:** enable or disable the fallback function 0—disable 1—enable<br>• **supportDRL:** true –Terminal support DRL for Visa, false—Terminel not support DRL for Visa.<br>• **enableBeeper:** true - enable beeper when card read successful , false - disable beeper when card read successful.<br>• **enableTapSwipeCollision:** true - endable Tap/Swipe card collision check, false - disable Tap/Swipe card collision check.<br>• **DisableCheckMSRFormat:** true – Disable check MSR format ,false – Check MSR format<br>• **NeedFallBackTryTimes:** (eg:1,2,3…) How many times does it take to fallback after the card reading fails.<br>• **DisableCheckMSRFormat:** true – check the magnetic stripe Format, false – Don`t |

check
Mag Stripe Format.

- **enableEncMagStripe:** true – Return Encrypt magnetic stripe, false – Return Plaintext magnetic stripe
- **MSRKeyIndex:** (1-4) DUKPT key index for encrypted magnetic stripe
- **forceInputPIN:** true - Force input of online PIN, valid for contactless. Note: This function cannot be used for L2 authentication

## 1.02 abortKernel

| Declare: | public void abortKernel () |
|---|---|
| Descriptio: | abort Kernel, stop check card, stop see phone check card |
| Parameter: | none |
| Return: | |
| Remark: | |

## 1.03 setAmountEx

| Declare: | **public Boolean setAmountEx(String amount, String cashbackAmount)** |
|---|---|
| Description: | Set the amount, cashback amount in response to onRequestSetAmount.<br>This function can be called before a transaction. The API will store the values and<br>If the function return False, the transaction should be terminate |
| Parameters: | • **amount** : The transaction amount.<br>• **cashbackAmount** : The cashback amount. |
| Return: | True - Amount values are set.<br>False - Failed to set values. |
| Remark: | Only use in **onRequestSetAmount** |

## 1.04 sendConfirmCardnoResult

| Declare: | **void sendConfirmCardnoResult(Boolean isconfirm)** |
|---|---|

| | |
|---|---|
| Description： | Confirm the card number is right,call at onRequestConfirmCardno() |
| Parameters： | True false |
| Return： | None |
| Remark： | None |

## 1.05 selectApplication

| | |
|---|---|
| Declare： | void selectApplication(int index) |
| Description： | An chip card may support multiple payment applications. The list of Applications<br>IDs supported by the card and device is returned in onRequestSelectApplication.<br>The App should prompt the customer for an application to continue the<br>transaction. |
| Parameters： | The index of the application selected. |
| Return： | None |
| Remark： | onRequestSelectApplication |

## 1.06 sendFinalConfirmResult

| | |
|---|---|
| Declare： | void sendFinalConfirmResult(boolean isConfirmed) |
| Description： | Send the final confirmation to proceed or cancel the transaction in response to onRequestFinalConfirm. |
| Parameters： | • **isConfirmed**: Indicate the confirmation result. True - Confirm to continue the transaction. False - Cancel the transaction. |
| Return： | None |
| Remark： | onRequestFinalConfirm |

## 1.07 sendOnlineProcessResult

| | |
|---|---|
| Declare： | void sendOnlineProcessResult(Boolean bOnlineResult, String tlv) |
| Description： | Send back the online process result to the card in response to onRequestOnlineProcess. To terminate the transaction, send null or empty string as result.<br>bOnlineResult<br>• true: go online success<br>• false: go online fail(unable to go online) |

| | |
|---|---|
| Parameters: | A set of transaction results in Tag-Length-Value format.<br>The following tags are usually returned to the card:<br>• Tag 8A: Authorisation Response Code (Mandatory)<br>• Tag 89: Authorisation Code<br>• Tag 91: Issuer Authentication Data<br>• Tag 71: Issuer Script Template 1<br>• Tag 72: Issuer Script Template 2<br>Example:<br>String tlv = “8A023030910831323334353637 38” |
| Return: | None |
| Remark: | onRequestOnlineProcess |

## 1.08 ProcOnlinePinAgain

| | |
|---|---|
| Declare: | public void ProcOnlinePinAgain() |
| Description: | If you not support bypass ,you can call this api ,restart pop up pinpad |
| Parameters: | None |
| Return: | None |
| Remark: | Use in online pin |

## 1.09 sendPinEntry

| | |
|---|---|
| Declare: | void sendPinEntry () |
| Description: | Tell User has entered online pin |
| Parameters: | |
| Return: | None |
| Remark: | Use in online pin |

## 1.10 bypassPinEntry

| | |
|---|---|
| Declare: | void bypassPinEntry() |
| Description: | Bypass the PIN entry step in response to onRequestPinEntry If the card does not accept bypassing, the transaction will be aborted. |
| Parameters: | None |
| Return: | None |
| Remark: | Use in online pin |

## 1.11 cancelPinEntry

| | |
|---|---|
| Declare： | void cancelPinEntry () |
| Description： | Cancel the PIN entry step in response to onRequestPinEntry If the card does not accept bypassing, the transaction will be aborted. |
| Parameters： | None |
| Return： | None |
| Remark： | Use in online pin |

## 1.12 updateTerminalParamters

| | |
|---|---|
| Declare： | Boolean updateTerminalParamters (ContantPara.CardSlot cardSlot, String tlv) |
| Description： | Update a terminal configuration parameter, provided that the tag is supported and allow to update. |
| Parameters： | ContantPara.CardSlot.ICC: Update ICC Terminal Paramters<br>ContantPara.CardSlot.PICC:Update PICC Terminal Paramters<br>ContantPara.CardSlot.UNKNOW: Update ICC and PICC Terminal Paramters.<br><br>A terminal configuration parameter in Tag-Length-Value format. See below table for the list of tags. |
| Return： | True – update success, False- update failed |
| Remark： | |

| Tag | Format | Length | describe |
|---|---|---|---|
| 9F1A | n3 | 2 bytes | Terminal Country Code. |
| 9F01 | n12 | 6 bytes | Transaction Currency Code. |
| 5F36 | b | 1 byte | Transaction Currency Exponent. |
| 9F4E | ans0-40 | Variable 0-40 | Merchant Name and Location. |
| 9F16 | Ans0-15 | Variable 0-15 bytes | Merchant Identifier. |
| 9F1C | An0-8 | Variable 0-8 bytes | Terminal Identification. |
| 9F33 | b | 3 bytes | Terminal Capabilities. |

| | | | |
|---|---|---|---|
| | | | (default: E0F8C8) |
| 9F35 | n2 | 1 byte | Terminal Type. (default: 22) |
| 9F40 | b | 5 byte | Additional Terminal Capabilities |
| 9F15 | n4 | 2 byte | Merchant Category Code |
| DF02 | b | 1 byte | Ramdom Transaction Switch |
| DF03 | b | 1 byte | Exception File Check Switch |
| DF04 | b | 1 byte | Support SM |
| DF05 | b | 1 byte | Valocity Check enable |
| DF7F | b | 5-17 byte | Terminals prioritize AID (PURE/VCCS kernel) |

## 1.13 updateCAPK

| | |
|---|---|
| Declare： | Boolean updateCAPK(ContantPara.Operation operation, Hashtable<String,String> capkParams) |
| Description： | Update a Certificate Authority Public Key object in the device. The CAPK object to be updated. The object contains the location of the CAPK in the device, the RID, index, modulus, exponent and the checksum of the CAPK, all data in hex string format.<br>**Rid**: Identify public key of certification center<br>**Index**: Identify public key of certification center together with RID<br>**Exponent**: Public key exponent<br>**Modulus**: Module value of public key<br>**Checksum**: To verify public key of certification center |
| Parameters： | To remove a CAPK entry at the specified location, set with the following values:<br>• RID (5 bytes) = "A000000004"<br>• Index (1 byte) = "01"<br>• Exponent (1 byte or 3 bytes) = "03"<br>• Modulus (N bytes) = "00"<br>• Size (2 bytes)  (not use)<br>• Checksum (20 bytes) = SHA-1 [RID \|\| Index \|\| Modulus \|\| Exponent] |
| Return： | True—update success  False—update failed |

| | |
|---|---|
| Remark： | Mandatory: RID&Index&Exponent&Modulus |

## 1.14 updateAID

| | |
|---|---|
| Declare： | boolean updateAID(ContantPara.Operation operation, Hashtable<String,String> aidParams) |
| Description： | Update the AID info.<br>modifying the AID info with updateAID command. |

| | |
|---|---|
| Parameters： | Hashtable keys<br>• **CardType:** “IcCard”<br>• **aid**: Application Identifier (AID).<br>• **appVersion**: Application Version Number.<br>• **contactTACDefault:** Identify the Acquirer’s conditions to reject transactions when the transaction is able to be completed online but the terminal lacks such capability.<br>• **contactTACDenial:** Identify the Acquirer’s conditions for not trying any online processing i.e. rejection of transactions.<br>• **contactTACOnline:** Identify the Acquirer’s conditions for online transactions<br><br>• **defaultTDOL**: Transaction Certificate Data Object List (TDOL).<br>• **defaultDDOL**: Dynamic Data Authentication Data Object List (DDOL). DDOL used for establishment of internal certification command when there is no DDOL in the card.<br>• **terminalFloorLimit**: Contact Terminal Floor Limit. The lowest offline limit allowed at the terminal in IC card purchase<br>•**AppSelIndicator:**application select indicator 00-party match ,01-full match ,default 00<br>• **AcquirerIdentifier**: Acquirer Identifier.<br>• **TerminalCapabilities**: terminal Capabilities 9F33<br>• **terminalCountryCode**: terminal country code 9F1A<br>• **ThresholdValue**:Value used in terminal risk management for random transaction selection. Present if the Combination supports Random Transaction Selection (EMV Mode only)<br>**Eg: "000000002000"**<br>• **TargetPercentage:** Present if the Combination supports Random Transaction Selection (EMV Mode only)<br>**Eg: "00"**<br>• **MaxTargetPercentage:**Value used in terminal risk management for random transaction selection - present if the Combination supports Random Transaction Selection (EMV Mode only)<br>**Eg: "00"**<br><br>Hashtable Values<br>• Data in hex string format.<br><br>**Example**:<br>`Hashtable<String, String> data = new Hashtable<~>();`<br>`data.put("CardType", "IcCard");`<br>`data.put("aid", "A0000000041010");` |

| | |
|---|---|
| Return： | True –update success　False—update failed |
| Remark： | |

## 1.15 updateExceptFile

| | |
|---|---|
| Declare： | boolean updateExceptFile(int mode,String ExceptTLV) |
| Description： | **Update black list to kernel** |
| Parameters： | **Mode:0—Add**<br>**1—Delete**<br>**2—Modify**<br>**3—Clear**<br>**ExceptTLV:**<br>**tag5A +tag5F34**<br>**eg: 5A0852364979362037965F340101** |
| Return： | None |
| Remark： | If you want not support Except file check, don`t update any black list |

## 1.16 LogOutEnable

| | |
|---|---|
| Declare： | void LogOutEnable(int enable) |
| Description： | **Enable or disable contact/contactless transaction log** |
| Parameters： | **0-disable**<br>**1-enable** |
| Return： | None |
| Remark： | adb pull /sdcard/UROPE/Trace.txt ….<br>adb pull /sdcard/UROPE/TraceCL.txt ….<br>adb pull /sdcard/UROPE/ICCMD.log … |

## 1.17 sendOfflinePINVerifyResult

| | |
|---|---|
| Declare： | boolean sendOfflinePINVerifyResult(int iResult) |
| Description： | **Send offline pin verify result to kernel (this api use in Androd os 8.0,other version not use this ,offline pin verify by kernel)** |
| Parameters： | **(-198)　//Return code error**<br>**(-202)　//IC command failed**<br>**(-192)　//PIN BLOCKED**<br>**(-199)　//user cancel or Pinpad timeout**<br>**(1)　//bypass**<br>**(0)　//success** |
| Return： | 0-success |

| | |
|---|---|
| Remark: | Call this in callback onRequestOfflinePINVerify |

## 1.18 getIssuerScriptResult

| | |
|---|---|
| Declare: | boolean getIssuerScriptResult() |
| Description: | Get Script Result |
| Parameters: | none |
| Return: | 0-get success if success will callback onReturnIssuerScriptResult |
| Remark: | Call this in callback onReturnIssuerScriptResult |

## 1.19 getEmvAIDDetail

| | |
|---|---|
| Declare: | public List<EmvAidData> getEmvAIDDetail() |
| Description: | Return all Aid paramters ,see the EmvAidData.class |
| Parameters: | none |
| Return: | |
| Remark: | Do not call during the transaction |

## 1.20 getNfcAIDDetail

| | |
|---|---|
| Declare: | public List<ContactlessAidData> getNfcAIDDetail () |
| Description: | Return all Aid paramters ,see the ContactlessAidData.class |
| Parameters: | none |
| Return: | |
| Remark: | Do not call during the transaction |

## 1.21 getCAPKDetail

| | |
|---|---|
| Declare: | public List<CAPK> getCAPKDetail(ContantPara.CardSlot cardSlot) |
| Description: | Return all CAPK paramters ,see the CAPK.class |
| Parameters: | none |
| Return: | |
| Remark: | Do not call during the transaction |

## 1.22 getAIDList

| Declare: | public List<String> getAIDList(ContantPara.CardSlot cardSlot) |
|---|---|
| Description: | Return all AID |
| Parameters: | none |
| Return: | |
| Remark: | Do not call during the transaction |

## 1.23 getCAPKList

| Declare: | public List<Hashtable<String,String>> getCAPKList(ContantPara.CardSlot cardSlot) |
|---|---|
| Description: | Return all CAPK |
| Parameters: | Key: RID<br>Index |
| Return: | List |
| Remark: | Do not call during the transaction |

## 1.24 getTlvByTagLists

| Declare: | public String getTlvByTagLists(List<String> TagList) |
|---|---|
| Description: | Get the TLV string based on tag |
| Parameters: | TagList:All tags |
| Return: | TLV data string |
| Remark: | none |

## 1.25 getEMVLibVers

| Declare: | public String getEMVLibVers(ContantPara.CardSlot cardSlot) |
|---|---|
| Description: | Get emv so version |
| Parameters: | none |
| Return: | Version string |
| Remark: | none |

## 1.26 getEMVjarVers

| Declare: | public String getEMVjarVers() |
|---|---|
| Description: | Get emv jar version |
| Parameters: | none |
| Return: | Version string |
| Remark: | none |

## 1.27 getValByTag

| Declare: | public String getValByTag(int tag) |
|---|---|
| Description: | Get one contact or contactless tag value |
| Parameters: | tag (e.g. 0x57) |
| Return: | Return tag value with string |
| Remark: | none |

## 1.28 setContext

| Declare: | public void setContext(Context context) |
|---|---|
| Description: | Set context, so emv can call UI and pinpad handle offlie pin |
| Parameters: | |
| Return: | |
| Remark: | Use in android 5.1 or 6.0 |

## 1.29 SetTLV

| Declare: | public int SetTLV(int tag, byte[] value, int valueLen) |
|---|---|
| Description: | Set one contact tag data |
| Parameters: | tag (e.g. 0x95)<br>value: tag value hex<br>valueLen: length of value |
| Return: | 0: success<br>Other: failed |
| Remark: | **Only use for contact** |

## 1.30 getMstripFlag

| Declare: | public int getMstripFlag () |
|---|---|
| Description: | Check the contactless card is EMV mode or Magstrip mode<br>Note: VISA not support Mstrip mode |
| Parameters: | |
| Return: | 0 M chip(EMV mode)<br>1 M strip (Master/Amex) |
| Remark: | 4 JCB legacy mode<br>8 Discover ZIP mode |

## 1.31 CheckCardIsOut

| Declare: | public boolean CheckCardIsOut(long ms) |
|---|---|
| Description: | Check IC card is removed from device , but this api can`t call during transaction ， Otherwise, it will cause a kernel card reading conflict , |
| Parameters: | milliseconds |
| Return: | True:if card is removed during timeout False:card not removed |
| Remark: | call it at ContantPara.CheckCardResult.NEED_FALLBACK or ContantPara.CheckCardResult.NOT_ICC or ContantPara.TransactionResult.ICC_CARD_REMOVED |

## 1.32 CheckCardIsOutDuringTrans

| Declare: | public boolean CheckCardIsOutDuringTrans(long ms) |
|---|---|
| Description: | Check IC card is removed from device , this api can only call during Pop the PIN pad for contact transaction ， Otherwise, it will cause a kernel card reading conflict |
| Parameters: | milliseconds |
| Return: | True:if card is removed during timeout False:card not removed |
| Remark: | Use in input PIN flow ,when the card removed |

## 1.33 GetField55ForSAMA

| Declare: | public String GetField55ForSAMA() |
|---|---|
| Description: | Get filed55 for SAMA |
| Parameters: | null |
| Return: | TLV |
| Remark: | Call it at go online callback |

## 1.34 getDeviceType()

| Declare: | public int getDeviceType() |
|---|---|
| Description: | Get Device Type for Contactless |
| Parameters: | null |
| Return: | 0 – Real card , other- Mobile Phone |
| Remark: | Call it at go online callback, If the kernel cannot determine, the default real card |

## 1.34 getField55ForJIO()

| Declare: | public Hashtable<String,String> getField55ForJIO(int KeySetNum) |
|---|---|
| Description: | Get filed55 for Relience |
| Parameters: | KeySetNum -- DUKPT KeySetNum |
| Return: | Hashtable : KEY /VALUE : EMVDATA KSN TRACKDATA |
| Remark: | Call it at go online callback |

## 1.35 GetL1Version

| Declare: | public Hashtable<String,String> GetL1Version(ContantPara.CardSlot cardSlot) |
|---|---|
| Description: | Get L1 Version of device |
| Parameters: | cardSlot – ICC or PICC |
| Return: | Hashtable : KEY /VALUE : IFM Hardware Software PCD |
| Remark: | Subject to the L1 certificate |

## 1.36 getNFCLibVers

| Declare: | public String getNFCLibVers(byte KernelID) |
|---|---|
| Description: | Get Kernel name and Version |
| Parameters: | KernelID – 02-Master 03-Visa 04-Amex 05-JCB 06-Discover 0D-Rupay 2D-PURE |
| Return: | Name and Version |
| Remark: | Subject to the L2 certificate |

## 1.37 updateCAPK

| Declare: | Boolean updateCAPK(ContantPara.CardSlot cardSlot, ContantPara.Operation operation, Hashtable<String,String> |
|---|---|

| | |
|---|---|
| | capkParams) |
| Description： | Update a Certificate Authority Public Key object in the device. The CAPK object to be updated. The object contains the location of the CAPK in the device, the RID, index, modulus, exponent and the checksum of the CAPK, all data in hex string format. |
| Parameters： | **cardSlot**: ContantPara.CardSlot.ICC ,ContantPara.CardSlot.PICC or ContantPara.CardSlot.UNKNOWN<br><br>**Rid**: Identify public key of certification center<br>**Index**: Identify public key of certification center together with RID<br>**Exponent**: Public key exponent<br>**Modulus**: Module value of public key<br>**Checksum**: To verify public key of certification center<br><br>To remove a CAPK entry at the specified location, set with the following values:<br>• RID (5 bytes) = "A000000004"<br>• Index (1 byte) = "01"<br>• Exponent (1 byte or 3 bytes) = "03"<br>• Modulus (N bytes) = "00"<br>• Size (2 bytes) (not use)<br>• Checksum (20 bytes) = SHA-1 [RID \|\| Index \|\| Modulus \|\| Exponent] |
| Return： | True—update success False—update failed |
| Remark： | Mandatory: RID&Index&Exponent&Modulus |

## 1.38 updateContactAID_TLV

| | |
|---|---|
| Declare： | public boolean updateContactAID_TLV(String TLV) |
| Description： | Update Contact AID Parameters, Use TLV String |

| | |
|---|---|
| Parameters： | AID TLV<br>**Tag:**<br>**M:**<br>**9F06 : AID**<br>**9F09 : Terminal Application Version(eg:0001)**<br>**DF11: TAC-Default (eg: 0000000000)**<br>**DF12: TAC-Online (eg: DC4004F800)**<br>**DF13: TAG-Denial (eg: 0000000000)**<br>**DF14: Default DDOL (eg: 9F3704)**<br>**9F1B: Terminal Floor Limit (eg: 00000000)**<br>**O:**<br>**DF01: Application Select Indicator (eg:00-Partial Match 01-Full Match)**<br>**DF02:Terminal Application Priority (eg:00-0F)**<br>**DF22: Default TDOL (eg: 9F0206)**<br>**DF15: ThresholdValue (eg: 000000002000)**<br>**DF16: MaxTargetPercentage(eg:00)**<br>**DF17: TargetPercentage(eg:00)**<br>**9F01: Acquirer ID (eg: 303030313131)**<br>**9F33: Terminal Capabilities(eg: E0F8C8)**<br>**9F1A:Terminal Country Code(eg:0356)**<br>**DF23:Terminal Floor Limit Check (eg:01 or 00)**<br>**5F2A:Transaction Currency Code (eg:0156)**<br>**5F36:Transaction Currency Code Exponent(eg:02)**<br>**DF24: Application Default Label (eg: 4D617374657243617264)**<br>**9F16: Merchant ID (eg: 123456789012345123456789012345)**<br>**9F1C:Terminal ID(eg: 3132333435363738)**<br>**9F15: Merchant Category Code(eg: 1122)**<br>**9F40:Additional Terminal Capabilities(eg: F000F0A001)**<br>**9F35:Terminal Type(eg:22)**<br>**Note: M--- Mandatory O-- Optional**<br><br>**Eg:**<br>9F0607A00000000410109F09020002DF010100DF020100DF1105000<br>0000000DF13050010000000DF1205DC4004F800DF22039F0206DF1<br>4039F3704DF1506000000002000DF170100DF1601009F1B0400000<br>0009F01063030303131319F3303E0F8C89F1A020682DF230101<br>Funs.Find_TLV<br>Funs.AddChild_TLV<br>Funs.New_TLV<br>Funs.Remove_TLV<br>You can use these Apis to construct TLV |
| Return： | true false |
| Remark： | Pay attention to the data length and format |

## 1.39 updateContactlessAID_TLV

| Declare： | public boolean updateContactlessAID_TLV(ContantPara.NfcCardType nfcCardType, String TLV) |
| --- | --- |
| Description： | Update Contactless AID Parameters, Use TLV String |

| | |
|---|---|
| Parameters： | AID TLV<br>**Note: M--- Mandatory  O-- Optional**<br><br>**Tag:**<br>**M:**<br>**9F06:  ApplicationIdentifier**<br><br>**ContantPara.NfcCardType.MasterCard:**<br>**DF8120: TerminalActionCodesDefault**<br>**DF8121: TerminalActionCodesDenial**<br>**DF8122: TerminalActionCodesOnLine**<br>**DF8123：Floor Limit**<br>**DF8124: NoOnDeviceCVM**<br>**DF8125 : OnDeviceCVM**<br>**DF8126 : CVMRequiredLimit**<br>**DF811A: DefaultUDOL**<br>**9F1D: TerminalRiskManagement**<br>**DF811B: KernelConfiguration**<br>**DF8117: CardDataInputCapability**<br>**DF8118: CVMCapabilityPerCVMRequired**<br>**DF8119:  CVMCapabilityNoCVMRequired**<br>**DF811E: MagStripeCVMCapabilityCVMRequired**<br>**DF811F: SecurityCapability**<br>**DF812C: MagStripeCVMCapabilityPerNoCVMRequired**<br>**Eg:**<br>**9F0607A00000000410109F09020002DF81170160DF81180160DF811**<br>**90108DF811A039F6A04DF811B0130DF810C01029F6D020001DF81**<br>**1E0110DF812C0100DF8123060000000000000DF8124060000500000**<br>**00DF8125060000500000000DF8126060000000030000DF811F0108DF**<br>**812205FC50BCF800DF8121050000000000DF812005FC50BC80009**<br>**F1D086C7A000000000000**<br><br>**ContantPara.NfcCardType.VisaCard:**<br>**9F66: TerminalTransactionQualifiers**<br>**9F92810D: TransactionLimit**<br>**9F92810E: CvmRequiredLimit**<br>**9F92810F: FloorLimit**<br>**9F92810A: LimitSwitch**<br>**9F928102: ProRestrictionDisable**<br><br>**Eg:**<br>**9F0607A00000000310109F6604360040009F92810D0699999999999**<br>**99F92810F060000000000009F92810E060000005000009F92810A02**<br>**FE009F1B0400000000** |

| | |
|---|---|
| Return： | true false |
| Remark： | Pay attention to the data length and format |

## 1.40 updateCAPK_TLV

| | |
|---|---|
| Declare： | Public boolean updateCAPK_TLV(ContantPara.CardSlot cardSlot,String TLV) |
| Description： | Update CAPK Parameters, Use TLV String |
| Parameters： | cardSlot: ICC,PICC,UNKNOWN<br>**ICC—contact**<br>**PICC—contactless**<br>**UNKNOWN-contact and contactless**<br>**Tag:**<br>**M:**<br>**9F06 : RID**<br>**9F22 : Index**<br>**DF02: Modulus**<br>**DF04: Exponent**<br>**O:**<br>**DF03:CheckSum**<br>**DF05:ExpiredDate**<br>**DF07:AlgorithmIndicator**<br><br>**Note: M--- Mandatory O-- Optional**<br><br>**Eg:**<br>9F0605A0000000039F220108DF040103DF050420401231DF0281B0<br>D9FD6ED75D51D0E30664BD157023EAA1FFA871E4DA65672B863<br>D255E81E137A51DE4F72BCC9E44ACE12127F87E263D3AF9DD9C<br>F35CA4A7B01E907000BA85D24954C2FCA3074825DDD4C0C8F186<br>CB020F683E02F2DEAD3969133F06F7845166ACEB57CA0FC26034<br>45469811D293BFEFBAFAB57631B3DD91E796BF850A25012F1AE3<br>8F05AA5C4D6D03B1DC2E568612785938BBC9B3CD3A910C1DA55<br>A5A9218ACE0F7A21287752682F15832A678D6E1ED0BDF03130000<br>00000000000000000000000000000000<br>Funs.Find_TLV<br>Funs.AddChild_TLV<br>Funs.New_TLV<br>Funs.Remove_TLV<br>You can use these Apis to construct TLV |

| | |
|---|---|
| Return: | true false |
| Remark: | Pay attention to the data length and format |

## 1.41 exportLogFilesToExternalStorage

| | |
|---|---|
| Declare: | public static boolean exportLogFilesToExternalStorage(Context context) |
| Description: | Export log files from internal storage to external storage |
| Parameters: | context: used to get internal storage path |
| Return: | true false |
| Remark: | used to get internal storage path |

## 1.42 getNfcCvmCode

| | |
|---|---|
| Declare: | public int getNfcCvmCode() |
| Description: | Gets the CVM code from the current NFC transaction |
| Parameters: | null |
| Return: | An integer code indicating the required verification method.<br>● 0x00-No verification required (NO CVM)<br>● 0x10- Signature verification required (SIGNATURE)<br>● 0x20- Online PIN verification required (ONLINE PIN)<br>● 0x30- Confirmation code has been verified (CONFIRMATION CODE VERIFIED)<br>● 0xF0- Not Applicable (N/A) |
| Remark: | None |

## 1.43 getODAstatus

| | |
|---|---|
| Declare: | public byte getODAstatus() |
| Description: | Gets the ODA result of the transaction |
| Parameters: | null |
| Return: | ● 0x01 SDA Fail<br>● 0x02 DDA Fail<br>● 0x03 CDA Fail<br>● 0x04 FDDA Fail<br><br>● 0x81 SDA SUCESS<br>● 0x82 DDA SUCESS<br>● 0x83 CDA SUCESS<br>● 0x84 FDDA SUCESS |

| | |
|---|---|
| Remark： | **only used for mastercard and visa card** |

### 1.44 setApproveRespcodeList

| | |
|---|---|
| Declare： | public boolean setApproveRespcodeList(String respcodeList) |
| Description： | Set the Authorization Response Code list that indicates an approved transaction from the host. |
| Parameters： | respcodeList：Authorisation Response Code List,Separate with'\|'. such as "00\|10\|11" |
| Return： | true or false |
| Remark： | **only used for contact trade.** |

# 2.Callback function

### 2.01 onRequestSetAmount

| | |
|---|---|
| Declare： | void onRequestSetAmount() |
| Description： | Request input of amount in the EMV process. If amount is input at startEmv, this callback will not be triggered. To cancel the process, call cancelSetAmount. |
| Return： | None |

### 2.02 onRequestConfirmCardno

| | |
|---|---|
| Declare： | void onRequestConfirmCardno() |
| Description： | Confirm the Card number, should call sendConfirmCardnoResult(true) |
| Return： | |

### 2.03 onReturnCheckCardResult

| | |
|---|---|
| Declare： | void onReturnCheckCardResult(CheckCardResult checkCardResult, Hashtable<String, String> decodeData) |
| Description： | Return the results in response to a swiped, inserted or tapped card. |

| | |
|---|---|
| Return： | •See enum CheckCardResult<br>Key: StripInfo<br>CardNo |

## 2.04 onRequestSelectApplication

| | |
|---|---|
| Declare： | void onRequestSelectApplication(ArrayList<String> appList) |
| Description： | Request selection of an application from the returned list. To cancel the process, call cancelSelectApplication. |
| Return： | Return a list of applications supported by the mPOS device and the EMV chip card. The applications are sorted by priority, where index 0 is the highest priority. |

## 2.05 onRequestPinEntry

| | |
|---|---|
| Declare： | void onRequestPinEntry(PinEntrySource pinEntrySource) |
| Description： | Request PIN entry from the specified PinEntrySource. To cancel the process, call cancelPinEntry. This callback can be triggered by startKernel. |
| Return： | Indicate the PIN entry source, see *PinEntrySource* enumeration. |

## 2.06 onRequestFinalConfirm

| | |
|---|---|
| Declare： | void onRequestFinalConfirm() |
| Description： | The EMV process request for a final confirmation before calling the first generate AC command. To proceed or cancel the transaction, call sendFinalConfirmResult. |
| Return： | None |

## 2.07 onRequestOnlineProcess

| | |
|---|---|
| Declare： | void onRequestOnlineProcess(String tlv, String dataKsn) |
| Description： | Request the transaction go online. To proceed or terminate the transaction, call sendOnlineProcessResult. |

| | |
|---|---|
| Return： | A set of transaction data in Tag-Length-Value format that should be sent back to online server.<br>Tlv : TLV data<br>dataKsn: KSN for data encryption |

## 2.08 onReturnBatchData

| | |
|---|---|
| Declare： | void onReturnBatchData(String tlv) |
| Description： | Return the batch data after completion of an EMV transaction. |
| Return： | A set of transaction data in Tag-Length-Value format that should be sent back to server for processing.(Use in offline approve) |

## 2.09 onReturnTransactionResult

| | |
|---|---|
| Declare： | void onReturnTransactionResult(TransactionResult transResult) |
| Description： | Return the transaction result. |
| Return： | Indicate the transaction result, see TransactionResult enumeration. |

## 2.10 onRequestDisplayText

| | |
|---|---|
| Declare： | void onRequestDisplayText(DisplayText displayText) |
| Description： | Return the message to prompt customer. |
| Return： | Indicate the message to be displayed, see DisplayText enumeration. |

## 2.11 onRequestOfflinePinEntry

| | |
|---|---|
| Declare： | void onRequestOfflinePinEntry(ContantPara.PinEntrySource pinEntrySource, int PinTryCount); |
| Description： | Usually not use |
| Return： | |

## 2.12 onRequestOfflinePINVerify

| | |
|---|---|
| Declare： | void onRequestOfflinePINVerify(ContantPara.PinEntrySource pinEntrySource, int pinEntryType, Bundle bundle); |
| Description： | **Offline pin verify(use in Android os 8.0)** |
| Parameters： | pinEntrySource-see euam ContantPara.PinEntrySource |

| | |
|---|---|
| Return： | pinEntryType:0-----Offline PlainPin 1--- Offline encryptionPin<br>bundle :key: ModuleLen<br>Module<br>ExponentLen<br>Exponent |
| Remark： | Please check the sample demo app |

## 2.13 onReturnIssuerScriptResult

| | |
|---|---|
| Declare： | void onReturnIssuerScriptResult(ContantPara.IssuerScriptResult scriptResult, String tlv) |
| Description： | Return Script Result ,tag DF31 is scarpt result<br>Tlv:-field55 data |
| Return： | none |

## 2.14 onNFCrequestTipsConfirm

| | |
|---|---|
| Declare： | void onNFCrequestTipsConfirm(ContantPara.NfcTipMessageID messageID ,String msg); |
| Description： | Request application display massage. |
| Parameters： | messageID:check enum NfcTipMessageID<br><br>msg:<br>*PLS_REMOVE_CARD*,<br>*PLS_USE_CONTACT_IC_CARD*,//1<br>*NEED_SIGNATURE*,//2<br>*END_APPLICATION*,//3<br>*SEE_PHONE_REMOVE_AND_PRESENT_CARD*,//4<br>*DISPLAY_BALANCE*,//5<br>*CARD_READ_OK*,//6<br>*PLS_SECOND_TAP_CARD*,//7<br>*APPLICATION_BLOCKED*,//8<br>*TRY_AGAIN_RESENT_CARD*,//9<br>*USE_MAG_STRIPE*,//10<br>*INSERT_SWIPE_OR_TRY_ANOTHER_CARD*,//11<br>*TERMINATE*,<br>*CARD_ERROR*,<br>*PROCESSING_ERROR*,<br>*UNKNOW* |

| | |
|---|---|
| Remark： | Called at processPayPassKernel |

## 2.15 onNFCrequestOnline

| Declare： | void onNFCrequestOnline (Hashtable<String, String> hashtable); |
|---|---|
| Description： | When transaction need go online |
| Parameters： | Hashtable:<br>"KSN" : custom format ksn, usually not use<br>"TRACKDATA" : custom format data, usually not use<br>"EMVDATA" : contactless transaction TLV data |
| Return： | |
| Remark： | If 0x5A(PAN) is not returned, please Parsing tag 0x57 to get PAN, if 0x57 is not returned, please check whether it's a mstrip card. For mastercard mstrip the Track 2 Equivalent Data will be save at tag 0x 9f6b |

## 2.16 onNFCrequestImportPin

| Declare： | void onNFCrequestImportPin (int type, int lasttimeFlag, String amt); |
|---|---|
| Description： | Contactless request import PIN |
| Parameters： | Parameters Not used now<br>Call the password keyboard in this callback function, you may need to get the card number<br><br>You can get the card number through tag5A, if tag5A is empty, you can get the card number through tag57 |
| Return： | |
| Remark： | |

## 2.17 onNFCTransResult

| Declare： | void onNFCTransResult (ContantPara.NfcTransResult result); |
|---|---|
| Description： | Return the transaction result to application |

| | |
|---|---|
| Parameters： | Result:<br>Check enum NfcTransResult<br>ONLINE_APPROVAL,<br>DECLINE_ONLINE,<br>TERMINATE,<br>OFFLINE_APPROVAL,<br>OTHER_INTERFACES,<br>RETRY,<br>CARD_REMOVED,<br>ISSUER_SCRIPT_UPDATE_SUCCESSFUL, //8<br>ISSUER_SCRIPT_UPDATE_FAILED, //9<br>DECLINE_OFFLINE |
| Return： | None |
| Remark： | If the result is OFFLINE_APPROVAL, please call getNfcCvmCode() to get the CVM (Cardholder Verification Method) code. Then, If the return value of the CVM code is 0x10 (indicating signature required), then require a signature when printing the slip. |

## 2.18 onNFCErrorInfor

| | |
|---|---|
| Declare： | void onNFCErrorInfor (ContantPara.NfcErrMessageID errID, String strErrInfo); |
| Description： | Contactless transaction error callback function |
| Parameters： | errID:check enum NfcErrMessageID<br><br>strErrInfo:<br>ERR_LOAD_CALLBACK,<br>ICS_PARAM_NOT_FIND,<br>KERNEL_ERR,<br>ERR_PIN_LENTH,<br>ERR_MULT_CARD,<br>ERR_CHECK_CARD,<br>AID_PARAM_NOT_FIND,<br>CAPK_PARAM_NOT_FIND,<br>GET_KERNEL_DATA_FAILED,<br>QPBOC_FDDA_FAILED,<br>PURE_ELE_CASH_CARD_NOT_ALLOW_ONLINE_TRANS, |
| Return： | |

| Remark： | |
|---|---|

## 2.19 onReturnNfcCardData

| Declare： | public void onReturnNfcCardData(Hashtable<String, String> hashtable) |
|---|---|
| Description： | Return contactless transaction TLV data |
| Parameters： | Hashtable:<br>"KSN" : custom format ksn, usually not use<br>"TRACKDATA" : custom format data, usually not use<br>"EMVDATA" : contactless transaction TLV data<br>“QPBOCTYPE”:if not support QPBOC ,it is NULL |
| Remark： | If 0x5A(PAN) is not returned, please Parsing tag 0x57 to get PAN, if 0x57 is not returned, please check whether it’s a mstrip card. For mastercard mstrip the Track 2 Equivalent Data will be save at tag 0x 9F6B |

# 3.EMV Enum

## 3.01 EmvOption

```
public static enum EmvOption
   {
      START, START_WITH_FORCE_ONLINE
   }
```

## 3.02 CheckCardMode

```
public static enum CheckCardMode{
      SWIPE, INSERT, TAP, SWIPE_OR_INSERT, SWIPE_OR_TAP,
INSERT_OR_TAP,
      SWIPE_OR_INSERT_OR_TAP,
   }
```

## 3.03 CheckCardResult

```
public static enum CheckCardResult{
    NO_CARD, INSERTED_CARD, NOT_ICC, BAD_SWIPE, MSR,  USE_ICC_CARD,
    TAP_CARD_DETECTED,  NEED_FALLBACK, TIMEOUT,CANCEL,DEVICE_BUSY
}
```

| Checkcard result | Result description |
|---|---|
| NO_CARD | No card has been detected, please check CheckCardMode |
| INSERTED_CARD | ICC card has been inserted |
| NOT_ICC | ICC card reading failed, please try to reinsert card |
| BAD_SWIPE | Magnetic stripe card reading failed, please try to re-swipe the card |
| MSR | Magnetic stripe card reading success |
| USE_ICC_CARD | Please insert ICC card |
| TAP_CARD_DETECTED | Tap card has been detected |
| NEED_FALLBACK | Please Tap or Swipe Card |
| TIMEOUT | Timeout , please try to read card again |
| CANCEL | Card reading has been canceled, please try to read card again |
| DEVICE_BUSY | Card reader is busy , please try to read card again later |
| MULT_CARD | Multiple cards detected , Please tap one card |

## 3.04 TransactionResult

```
 public static enum TransactionResult
    {
        OFFLINE_APPROVAL,ONLINE_APPROVAL, TERMINATED,
OFFLINE_DECLINED,ONLINE_DECLINED, CANCELED,  CANCELED_OR_TIMEOUT,
        CARD_BLOCKED_APP_FAIL,  NO_EMV_APPS, ICC_CARD_REMOVED,
        SELECT_APP_FAIL, INVALID_ICC_DATA,
            APPLICATION_BLOCKED_APP_FAIL
    }
```

| Transaction result | Result description |
|---|---|
| OFFLINE_APPROVAL | Offline request has been approved |

| | |
|---|---|
| ONLINE_APPROVAL | Online request has been approved |
| TERMINATED | Transaction has been terminated |
| OFFLINE_DECLINED | Offline request has been rejected |
| ONLINE_DECLINED | Online request has been rejected |
| CANCELED | Transaction has been canceled |
| CANCELED_OR_TIMEOUT | Transaction has been canceled or timeout |
| CARD_BLOCKED_APP_FAIL | Card has been blocked |
| NO_EMV_APPS | Terminal does not support applications of card |
| ICC_CARD_REMOVED | ICC card has been removed |
| SELECT_APP_FAIL | Terminal has been fail to select application of card |
| INVALID_ICC_DATA | ICC card data is invalid |
| APPLICATION_BLOCKED_APP_FAIL | Application has been blocked |

## 3.05 DisplayText

```
public static enum DisplayText
   {

      USE_MAG_STRIPE,
       APPROVED_PLEASE_SIGN,//if CVM is Signature

   }
```

## 3.06 PinEntryResult

```
public static enum PinEntryResult{
       ENTERED, CANCEL, TIMEOUT, BYPASS, WRONG_PIN_LENGTH,
INCORRECT_PIN
   }
```

## 3.07 PinEntrySource

```
    public static enum PinEntrySource{
       PHONE, KEYPAD
   }
```

## 3.08 NfcTransResult

```
public static enum NfcTransResult{
      ONLINE_APPROVAL,
      DECLINE_ONLINE,
      TERMINATE,
      OFFLINE_APPROVAL,
      OTHER_INTERFACES,
      RETRY,
      CARD_REMOVED,
      ISSUER_SCRIPT_UPDATE_SUCCESSFUL, //8
      ISSUER_SCRIPT_UPDATE_FAILED, //9
      DECLINE_OFFLINE}
```

## 3.09 NfcTipMessageID

```
public static enum NfcTipMessageID
{
      PLS_REMOVE_CARD,
      PLS_USE_CONTACT_IC_CARD,//1
      NEED_SIGNATURE,//2
      END_APPLICATION,//3
      SEE_PHONE_REMOVE_AND_PRESENT_CARD,//4
      DISPLAY_BALANCE,//5
      CARD_READ_OK,//6
      PLS_SECOND_TAP_CARD,//7
      APPLICATION_BLOCKED,//8
      TRY_AGAIN_RESENT_CARD,//9
      USE_MAG_STRIPE,//10
      INSERT_SWIPE_OR_TRY_ANOTHER_CARD,//11
      TERMINATE,
      CARD_ERROR,
      PROCESSING_ERROR,
      UNKNOW
}
```

## 3.10 NfcErrMessageID

```
public static enum NfcErrMessageID
{
    ERR_LOAD_CALLBACK,//0
    ICS_PARAM_NOT_FIND,// need update terminal paramters
    KERNEL_ERR,
    ERR_PIN_LENTH,
    ERR_MULT_CARD,
    ERR_CHECK_CARD,
    AID_PARAM_NOT_FIND,
    CAPK_PARAM_NOT_FIND,
    GET_KERNEL_DATA_FAILED,
    QPBOC_APPLICATION,//9
    QPBOC_FDDA_FAILED,
    PURE_ELE_CASH_CARD_NOT_ALLOW_ONLINE_TRANS,
    UNKNOW
}
```

## 3.11 Operation

```
public static enum Operation
  {
    CLEAR,    // clear contact and contactless AID
    ADD,
    CLEAR_EMV_AID, // only clear contact AID
    CLEAR_NFC_AID //only clear contactless AID
  }
```

# 4.Appendix

## 4.1 CVM processing FAQ

### Q1: How to handle Online PIN verification?

**Trigger**: The EMV kernel invokes the *onRequestPinEntry()* callback when cardholder PIN verification is required.

**Application Response**: Call *getPinBlock()* or *getPinBlockEx()* (refer to device SDK) to launch the PIN entry interface.

**User Action Handling**:

- Success: Call *mEmvApi.sendPinEntry()* to confirm PIN submission.
- Cancellation: Call *mEmvApi.cancelPinEntry()* if the user aborts input.
- Skip (if allowed): Call *mEmvApi.bypassPinEntry()* for scenarios bypassing PIN validation.

### Q2: How does EMV the kernel handle offline plain Pin and offline encrypt pin?

- The EMV kernel handles PIN entry and verification automatically via the PIN pad.
- **No application intervention required** (Try test with sample application).

### Q2: How is cardholder signature triggered in EMV transactions?

The EMV kernel requests signature verification by invoking *onRequestDisplayText(ContantPara.DisplayText.APPROVED_PLEASE_SIGN)* with parameter *ContantPara.DisplayText.APPROVED_PLEASE_SIGN*

## 4.2 EMV kernel using FAQ

### Q1: Missing PAN (Tag 0x5A) – How to Fallback?

if 0x5A not return, need use 0x57 data to get PAN.

### Q2: How to Debug EMV (Contact/UPI) and NFC Issues?

1. **Enable Logging**: Call *LogOutEnable()* API(activates both EMV & NFC logs).
2. **Pull Logs Separately:** Pull logs via ADB.

```
# For EMV (Contact/UPI) issues
adb pull /data/data/com.urovo.demos/files/UROPE/Trace.txt

# For NFC (Contactless) issues
adb pull /data/data/com.urovo.demos/files/UROPE/TraceCL.txt
```

### Q3: Offline PIN fails on Android 5.1/6.0?

**Fix:** Call *setContext(MainActivity.this)* before transaction.

## 4.2 Contact

if you have any issue , please contact Xiray or Ron, xiaofeng.xia@urovo.com, an.luo@urovo.com , we will help you.